Ridout questions

Differential Equations (University of Melbourne)



Scan to open on Studocu

Downloaded by no no (hahakix253@homvela.com)
Studocu is not sponsored or endorsed by any college or university

Lecturer: David Ridout
School of Mathematics and Statistics

Second Semester, 2017
University of Melbourne

MAST20030: Differential Equations
A selection of exercises

These are (mostly) intended to be "easy" exercises, designed to make sure that you can perform the steps required to recognise and solve some classes of different equations. I've borrowed most of these exercises from various textbooks, so please have a look in the library if you find yourself desperate to do more, more, more!
And please let me know if you find that any of these questions are unclear, have errors, or appear to be unsolvable. It's very easy to make a typo that transforms a benign problem into a beast. I *strongly* urge you to check your solutions using a computer algebra package (or Wolfram alpha or . . .).

Question 1.

(a) Find the general solution of the following ODEs:

$$\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{y}{t} = 2 + t^2, \qquad t\frac{\mathrm{d}y}{\mathrm{d}t} + 2\frac{y}{t} = \mathrm{e}^t + \log(t),$$
$$(t^2+1)\frac{\mathrm{d}y}{\mathrm{d}t} - 2ty = t^2 + 1, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} + \tan(t)y = \sec(t),$$
$$t\frac{\mathrm{d}y}{\mathrm{d}t} + (t+1)y = 3\mathrm{e}^{-t}.$$

(b) Find the unique solution of the following initial value problems:

$$\frac{\mathrm{d}y}{\mathrm{d}t} + 2y = t + \sin(t), \quad y(0) = 4, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} + ty = t^3, \quad y(0) = 3,$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{4y}{t+2} = 3, \quad y(0) = -5, \qquad (2t^2+1)\frac{\mathrm{d}y}{\mathrm{d}t} - 10ty = 4t, \quad y(0) = 7,$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} + y = \mathrm{e}^{-2t}, \quad y(0) = 1.$$

In which intervals are the solutions valid?

Question 2.

(a) Find the general solutions of the following homogeneous ODEs:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 3y = 0, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 5\frac{\mathrm{d}y}{\mathrm{d}t} + 4y = 0,$$
$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} - 3y = 0, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 0.$$

Now find the general solutions to the following *in*homogeneous ODEs:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 3y = 3\mathrm{e}^t, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 5\frac{\mathrm{d}y}{\mathrm{d}t} + 4y = 10\sinh(t),$$
$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} - 3y = 2 + 3\mathrm{e}^t, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = \sin(t).$$

(b) By trialling solutions of the form $y = t^n$, find all solutions of the following ODEs:

$$t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 7t\frac{\mathrm{d}y}{\mathrm{d}t} + 9y = \log(t), \qquad t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + t\frac{\mathrm{d}y}{\mathrm{d}t} - y = t,$$
$$t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 4t\frac{\mathrm{d}y}{\mathrm{d}t} + 6y = t^4\mathrm{e}^t, \qquad 2t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 5t\frac{\mathrm{d}y}{\mathrm{d}t} + y = t(t-1).$$

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

(c) For the following ODEs, show that the function y_c is a complementary solution and use it to find the general solution:

$$y_c = t - 1, \qquad (2t - t^2)\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2(t-1)\frac{\mathrm{d}y}{\mathrm{d}t} - 2y = 1,$$

$$y_c = t, \qquad t^3\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + t\frac{\mathrm{d}y}{\mathrm{d}t} - y = t.$$

In both cases, find the unique solution satisfying $y(1) = 0$ and $y'(1) = 0$. What happens if we replace the initial conditions by $y(0) = a$ and $y'(0) = b$, for some $a, b \in \mathbb{R}$?

Question 3.

We study the inhomogeneous second-order linear ODE

$$t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + t\frac{\mathrm{d}y}{\mathrm{d}t} + y = 1.$$

(a) Show that $v(t) = \cos(\log t)$ is a complementary solution for $t > 0$.

(b) Use reduction of order to deduce the general solution to the inhomogeneous ODE above.

(c) Explain how trialling $y = t^n$ could have led directly to the complementary solutions.

Question 4.

Find the general solutions of the following higher-order linear ODEs:

$$y''' + 6y'' + 12y' + 8y = 0, \qquad ty''' - y'' = 0,$$

$$y'''' + 2y''' + y'' = 0, \qquad y'''' - 5y'' + 4y = 0,$$

$$y''' + y' = 6t, \qquad y''' + 3y'' + 3y' + y = 4\mathrm{e}^{-t}.$$

Don't forget to check that your complementary solutions are linearly independent!

Question 5.

(a) Show that the wronskian of *n linearly dependent* (sufficiently differentiable) functions vanishes.
Hint: write down the linear dependence relation, differentiate it sufficiently many times and rewrite the result in terms of the matrix whose determinant is the wronskian. What does the result say about this matrix?

(b) The contrapositive of the previous statement is: if the wronskian of n (sufficiently differentiable) functions is non-zero, then the functions are linearly independent. This is the test we have used to check if complementary solutions are linearly independent. Show that the converse to this statement is false by verifying that

$$y_1(t) = t^2 \quad \text{and} \quad y_2(t) = t|t|$$

are differentiable for all $t \in \mathbb{R}$, are linearly independent as functions on $\mathbb{R}$, and have vanishing wronskian.
This counterexample is due to Peano who also proved that the converse is true if the functions are analytic.

Question 6.

(a) Rewrite the following systems of ODEs as vector-valued first-order linear ODEs and find their general solutions:

$$\left\{\begin{array}{l} f'(t) - g'(t) - f(t) = \cos t, \\ g'(t) - 2f(t) = \mathrm{e}^t \end{array}\right\},$$

$$\left\{\begin{array}{l} f'(t) - 2g'(t) - g(t) = t^2, \\ f'(t) - 3g(t) = 0 \end{array}\right\},$$

$$\left\{\begin{array}{l} f'(t) = -f(t) + g(t), \\ g'(t) = 2f(t) - 2g(t) + 2h(t), \\ h'(t) = g(t) - h(t) \end{array}\right\}.$$

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

(b) Find the unique solution of the following IVPs in whatever manner you deem fit:

$$\left\{\begin{array}{ll} f'(t) = -2f(t) + g(t), & f(0) = 3, \\ g'(t) = -3f(t) + 2g(t) + 2\sin t, & g(0) = 4 \end{array}\right\},$$

$$\left\{\begin{array}{ll} f'(t) = -f(t) - 3g(t) + 2h(t), & f(0) = -3, \\ g'(t) = -2g(t), & g(0) = 0, \\ h'(t) = -g(t) - 2h(t), & h(0) = 3 \end{array}\right\}.$$

(c) Find the solution to the following IVPs using a matrix exponential:

$$\left\{\begin{array}{ll} f'(t) = -2f(t) + g(t), & f(0) = a, \\ g'(t) = -4f(t) + 3g(t), & g(0) = b \end{array}\right\},$$

$$\left\{\begin{array}{ll} f'(t) = -2g(t), & f(0) = a, \\ g'(t) = 2f(t), & g(0) = b \end{array}\right\},$$

$$\left\{\begin{array}{ll} f'(t) = 3f(t) + g(t) - 2h(t), & f(0) = a, \\ g'(t) = -f(t) + 2g(t) + h(t), & g(0) = b, \\ h'(t) = 4f(t) + g(t) - 3h(t), & h(0) = c \end{array}\right\}.$$

Question 7.

Use Taylor series about $t = 0$ to find as many solutions as you can to the following ODEs. If possible, give a closed form solution for each and determine the radius of convergence. If not, give the first six non-zero terms of the series.

$$(1-t)y''(t) + y(t) = 0, \qquad (2+t^2)y''(t) - ty'(t) + 4y(t) = 0,$$
$$ty''(t) + y'(t) + ty(t) = 0, \qquad (1+t^2)y''(t) - 4ty'(t) + 6y(t) = 0,$$
$$(4-t^2)y''(t) + 2y(t) = 0, \qquad 2y''(t) + ty'(t) + 3y(t) = 0,$$
$$y''(t) + ty'(t) + y(t) = 0, \qquad y''(t) - ty'(t) - t^2y(t) = 0,$$
$$ty''(t) + 2y'(t) + ty(t) = 0, \qquad ty''(t) + 3y'(t) - y(t) = 0.$$

Question 8.

In lectures, we studied an example of a Bessel equation $ty''(t) + y'(t) + ty(t) = 0$ and showed that it has only one Taylor series solution about $t = 0$ (up to multiplying by a constant):

$$y_1(t) = \sum_{m=0}^{\infty} \frac{(-1)^m t^{2m}}{4^m (m!)^2}.$$

Moreover, the radius of convergence was shown to be infinite.

(a) Compute the wronskian of y_1 and another linearly independent solution y_2 and use this to show that y_2 diverges like $\log t$ for $t \to 0^+$. [We checked this using reduction of order in lectures, it's really the same!]

(b) Your wronskian analysis should naturally lead you to a generalised series expansion for $y_2(t)$ about $t = 0$. Find the first three non-vanishing terms in this series.

(c) Explain why this analysis suggests that y_2 may be expressed as a series of the form

$$y_2(t) = \log t\, y_1(t) + \sum_{j=0}^{\infty} c_j t^j.$$

Compute the c_j and determine the t for which $y_2(t)$ converges.

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

Question 9.

(a) Determine whether the following integrals converge or diverge:

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}t}{(1+t^2)^5}, \qquad \int_1^{\infty} \frac{\mathrm{e}^t}{t^7}\,\mathrm{d}t, \qquad \int_0^{\infty} \mathrm{e}^{-t}\sin t\,\mathrm{d}t, \qquad \int_0^{\infty} t\mathrm{e}^{-t}\log t\,\mathrm{d}t.$$

(b) Compute the Laplace transforms of the following functions and indicate the domain of the transformed function:

$$\mathrm{e}^{-5t}, \qquad \cos(2t), \qquad -3\sin(4t)+7\mathrm{e}^{3t}, \qquad \sqrt{t}+\frac{1}{\sqrt{t}}, \qquad \mathrm{e}^t\sin t, \qquad \frac{\cosh(2t)}{\sqrt{t}}, \qquad 2\sinh(3t)\sin(3t).$$

(c) Determine the inverse Laplace transform of the following functions, all assumed to be defined for $\operatorname{Re} s$ sufficiently large:

$$\frac{1}{(s-2)(s^2+1)}, \qquad \frac{1}{s^2(s^2+1)}, \qquad \frac{1}{s^3-8}, \qquad \frac{3s-8}{s^2-5s+6}, \qquad \frac{s^2+20s+9}{(s-1)^2(s^2+9)}, \qquad \frac{3s-2}{s^2-2s+10}.$$

Question 10.

(a) Rewrite the following functions in piecewise-defined form (*ie.*, don't use Heavisides):

$$H(t)+3H_1(t)-H_2(t), \quad t^2H_2(t), \quad \sin(2t-\tfrac{2\pi}{3})H_{\pi/3}(t), \quad (t-5)^2(1-2H_4(t)), \quad \sin(H(t)), \quad H(\sin(t)).$$

(b) Compute the Laplace transforms of the following piecewise-defined functions and indicate the domain of the transformed function:

$$f(t)=\begin{cases} t & \text{if } 0\leqslant t\leqslant 1,\\ 1 & \text{if } t>1,\end{cases} \qquad g(t)=\begin{cases}\sin t & \text{if } 0\leqslant t\leqslant \pi,\\ 0 & \text{if } t>\pi,\end{cases}$$

$$h(t)=\begin{cases} t & \text{if } 0\leqslant t\leqslant 1,\\ 2-t & \text{if } 1<t\leqslant 2,\\ 0 & \text{if } t>2,\end{cases} \qquad j(t)=\begin{cases}1 & \text{if } 0\leqslant t\leqslant 2,\\ 0 & \text{if } 2<t<3,\\ -1 & \text{if } 3\leqslant t\leqslant 6,\\ 0 & \text{if } t>6.\end{cases}$$

(c) Find the inverse Laplace transforms of the following functions, all assumed to be defined for $\operatorname{Re} s$ sufficiently large:

$$\frac{5\mathrm{e}^{-s}}{s^3}, \qquad \frac{\mathrm{e}^{-2s}}{s+4}, \qquad \frac{s^2}{s^2+4}, \qquad \frac{s\mathrm{e}^{-3s}}{s^2+9}, \qquad \frac{4\mathrm{e}^{-\pi s/2}}{s^2(s^2+9)}, \qquad \frac{s^3-s^2-s+37}{s^3+s-10}, \qquad \frac{6s\mathrm{e}^{-2s}}{(s^2+\pi^2)^2}.$$

Question 11.

(a) Use Laplace transforms to solve the following IVPs:

$$\begin{aligned}
y''(t)+2y'(t)-3y(t)&=32\mathrm{e}^{5t}, && y(0)=0, && y'(0)=12,\\
y''(t)+3y'(t)-4y(t)&=0, && y(0)=0, && y'(0)=1,\\
y''(t)-y'(t)-6y(t)&=\mathrm{e}^{-2t}, && y(0)=2, && y'(0)=3,\\
y''(t)+9y(t)&=6\cos(3t), && y(0)=1, && y'(0)=6,\\
y''(t)-2y'(t)+2y(t)&=2t-1, && y(0)=\tfrac{1}{2}, && y'(0)=3,\\
y''(t)+2y'(t)+y(t)&=4\sin t, && y(0)=-2, && y'(0)=1,\\
y''(t)+4y'(t)+5y(t)&=25t, && y(0)=-5, && y'(0)=7,\\
y'''(t)-y''(t)+9y'(t)-9y(t)&=0, && y(0)=0, && y'(0)=3 \quad y''(0)=6,\\
y'''(t)+2y''(t)+y'(t)+2y(t)&=2, && y(0)=3, && y'(0)=-2, \quad y''(0)=3,\\
y''''(t)-2y''(t)+y(t)&=8\mathrm{e}^t, && y(0)=1, && y'(0)=2, \quad y''(0)=1, \quad y'''(0)=10.
\end{aligned}$$

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

(b) And these ones, where $f(t) = \begin{cases} 1 & \text{if } t < 1, \\ 0 & \text{if } t > 1 \end{cases}$:

$$\begin{aligned}
y'(t) + y(t) &= f(t), & y(0) &= 0, \\
y''(t) + 2y'(t) + y(t) &= f(t), & y(0) &= 1, & y'(0) &= 0, \\
y''(t) + 4y(t) &= (1 - H_{2\pi}(t)) \cos t, & y(0) &= 0, & y'(0) &= 0, \\
y''(t) + 4y(t) &= f(t), & y(0) &= 0, & y'(0) &= 0, \\
y''(t) + 4y(t) &= f(t), & y(0) &= 1, & y'(0) &= 1, \\
4y''(t) + 9y(t) &= 5\delta_2(t), & y(0) &= 2, & y'(0) &= 0, \\
y''(t) + 4y'(t) + 5y(t) &= 5\mathrm{e}^{-2t} + 3\delta_3(t) + 2H_4(t), & y(0) &= 0, & y'(0) &= 0, \\
y''(t) + 6y'(t) + 9(t) &= 2\mathrm{e}^{-3t} + 4\delta_2(t) + 4\delta_4(t) + 4\delta_6(t), & y(0) &= 0, & y'(0) &= 0.
\end{aligned}$$

(c) And (finally) these ones, where g is left unspecified:

$$\begin{aligned}
y''(t) + \omega^2 y(t) &= g(t), & y(0) &= 1, & y'(0) &= 0, \\
y''(t) + 3y'(t) + 2y(t) &= g(t), & y(0) &= 0, & y'(0) &= 0, \\
y''(t) + 2y'(t) + y(t) &= g(t), & y(0) &= 1, & y'(0) &= 1.
\end{aligned}$$

Question 12.

Laplace transforms can also be used to solve systems of linear ODEs. Let $\boldsymbol{y}(t)$ be a vector-valued function whose components all have Laplace transforms. Let $(\mathscr{L}\boldsymbol{y})(s)$ denote the vector-valued function (of s) obtained by applying the Laplace transform to each component of $\boldsymbol{y}(t)$ individually.

(a) Show, by considering components, that the vector-valued first-order linear IVP

$$\boldsymbol{y}'(t) + \mathsf{A}\boldsymbol{y}(t) = \boldsymbol{b}(t), \qquad \boldsymbol{y}(0) = \boldsymbol{y}_0,$$

where A does not depend on t, may be transformed into

$$(\mathscr{L}\boldsymbol{y})(s) = (s\mathsf{I} + \mathsf{A})^{-1}(\boldsymbol{y}_0 + (\mathscr{L}\boldsymbol{b})(s)).$$

Why does the right-hand side make sense for sufficiently large s?

(b) Solve the following vector-valued first-order linear IVPs using Laplace transforms:

$$\begin{aligned}
\boldsymbol{y}'(t) + \begin{pmatrix} 1 & 1 \\ -3 & 5 \end{pmatrix} \boldsymbol{y}(t) &= \boldsymbol{0}, & \boldsymbol{y}(0) &= \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \\
\boldsymbol{y}'(t) + \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix} \boldsymbol{y}(t) &= \begin{pmatrix} 0 \\ 2\mathrm{e}^{t} \end{pmatrix}, & \boldsymbol{y}(0) &= \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \\
\boldsymbol{y}'(t) + \begin{pmatrix} 1 & 1 \\ -5 & -1 \end{pmatrix} \boldsymbol{y}(t) &= \begin{pmatrix} 0 \\ -4 \end{pmatrix}, & \boldsymbol{y}(0) &= \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \\
\boldsymbol{y}'(t) + \begin{pmatrix} 2 & 1 \\ -2 & 0 \end{pmatrix} \boldsymbol{y}(t) &= \begin{pmatrix} \mathrm{e}^{-t} \\ -\mathrm{e}^{-t} \end{pmatrix}, & \boldsymbol{y}(0) &= \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \\
\boldsymbol{y}'(t) + \begin{pmatrix} 1 & 3 & -3 \\ 0 & 4 & -3 \\ 0 & 6 & -5 \end{pmatrix} \boldsymbol{y}(t) &= \boldsymbol{0}, & \boldsymbol{y}(0) &= \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \\
\boldsymbol{y}'(t) + \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 1 & 2 & 5 \end{pmatrix} \boldsymbol{y}(t) &= \begin{pmatrix} 0 \\ 0 \\ 36 \end{pmatrix}, & \boldsymbol{y}(0) &= \boldsymbol{0}.
\end{aligned}$$

(c) For each of the matrices A appearing in the previous part, compute the inverse Laplace transform of $(s\mathsf{I} + \mathsf{A})^{-1}$ by applying $\mathscr{L}^{-1}$ to each entry individually. Now compute the matrix exponential $\mathrm{e}^{-\mathsf{A}t}$. Gasp in amazement.

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

Question 13.

In lectures, we stated the eigenvalues and eigenfunctions for the following BVP with periodic boundary conditions (with $L > 0$):

$$y''(x) + \lambda y(x) = 0, \qquad y(0) = y(L), \quad y'(0) = y'(L).$$

(a) Check that the eigenvalues are as stated in lectures, being careful about zero (as always).

(b) Check that the eigenfunctions are as stated in lectures, being careful about linear independence.

(c) Check that the eigenfunctions given in lectures form an orthonormal set with respect to the inner product

$$\langle f, g\rangle = \frac{1}{L}\int_{-L}^{L} f(x)^* g(x)\,\mathrm{d}x,$$

where the functions f and g satisfy periodic boundary conditions on $[-L, L]$.

(d) Check that $\mathrm{i}\dfrac{\mathrm{d}}{\mathrm{d}x}$ and $-\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$ are self-adjoint with respect to this inner product and these boundary conditions.

(e) If a function f on $[-L, L]$ may be written in the form

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\left[a_n \cos\frac{\pi n x}{L} + b_n \sin\frac{\pi n x}{L}\right],$$

derive the formulae for a_0 and the a_n and b_n given in lectures.

Question 14.

Consider the following BVPs with "mixed" boundary conditions:

$$y''(x) + \lambda y(x) = 0, \qquad y(0) = y'(L) = 0 \quad (L > 0).$$

(a) Prove that the eigenvalues and eigenfunctions of the BVP are

$$\lambda_n = \frac{\pi^2(n - \frac{1}{2})^2}{L^2}, \quad y_n(x) = \sin\frac{\pi(n - \frac{1}{2})x}{L}; \qquad n = 1, 2, 3, \ldots,$$

as stated in lectures.

(b) The eigenfunctions $y_n(x)$ are guaranteed to be orthogonal with respect to the inner product

$$\langle f, g\rangle = \alpha_L \int_{-L}^{L} f(x)^* g(x)\,\mathrm{d}x,$$

where α_L is an arbitrary positive constant. Determine α_L so that the eigenfunctions are orthonormal.

(c) Suppose that a function f on $[0, L]$ may be written as a linear combination of eigenfunctions:

$$f(x) = \sum_{n=1}^{\infty} b_n y_n(x).$$

Use orthonormality to derive a formula for the b_n.

(d) Repeat the previous three parts for the "mixed" BVP

$$y''(x) + \lambda y(x) = 0, \qquad y'(0) = y(L) = 0 \quad (L > 0).$$

Question 15.

Consider the following family of BVPs:

$$y''(x) + \lambda y(x) = 0, \qquad y(0) = 0, \quad y'(L) = \alpha y(L) \qquad (\alpha \in \mathbb{R} \setminus \{0\}).$$

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

(a) Check if $\mathrm{i}\dfrac{\mathrm{d}}{\mathrm{d}x}$ and $-\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$ are self-adjoint with respect to the inner product

$$\langle f, g\rangle = \int_0^L f(x)^* g(x)\,\mathrm{d}x$$

and these boundary conditions. What can you conclude about the eigenvalues and eigenfunctions of this BVP from your answer?

(b) Analyse the eigenvalues of this BVP. For which α is 0 an eigenvalue? For which α do you have a negative eigenvalue? You will not be able to compute the eigenvalues exactly, in general, but you should describe how many eigenvalues of each sign there are, for each α.

(c) Compute the eigenfunctions for all eigenvalues. You should find that the answers can be written in terms of the unknown eigenvalues.

(d) Set $\alpha L = 1$. Write down the smallest eigenvalue exactly and use a computer to determine the next three smallest numerically.

(e) Check that the eigenfunctions for $\alpha L = 1$ form an orthogonal set with respect to the given inner product. [This is a little tricky to show directly, though you should give it a shot. Is your answer to the first part confirmed?]

Question 16.

Compute the Fourier series of each of the following functions, extended periodically from the domains indicated, and determine the function that it (pointwise) converges to. In each case, is the convergence uniform?

$$f(x) = H(x), \qquad -L < x < L$$

$$f(x) = \begin{cases} -L - x & \text{if } -L < x < 0, \\ L - x & \text{if } 0 < x < L, \end{cases}$$

$$f(x) = \begin{cases} -1 & \text{if } -2 < x < 0, \\ 1 & \text{if } 0 < x < 2, \end{cases}$$

$$f(x) = \begin{cases} 0 & \text{if } 1 < |x| < 2, \\ x & \text{if } |x| < 1, \end{cases}$$

$$f(x) = \frac{\mathrm{e}^{-|x|/\varepsilon}}{2\varepsilon}, \quad -1 < x < 1$$

$$f(x) = \begin{cases} -\pi - x & \text{if } -\pi < x < -\frac{\pi}{2}, \\ x & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2}, \\ \pi - x & \text{if } \frac{\pi}{2} < x < \pi. \end{cases}$$

Question 17.

Recall that a piecewise-continuous *real*-valued function f on $[-L, L]$ has a real Fourier series, with Fourier coefficients a_n and b_n, as well as a complex Fourier series, with coefficients c_n. Moreover, both these series converge to $f(x)$ for all $-L < x < L$ (except perhaps for a finite set of points where f is discontinuous).

(a) Determine the relationship between the a_n, b_n and c_n.

(b) Substitute the *complex* Fourier series of f into $\displaystyle\int_{-L}^{L} |f(x)|^2\,\mathrm{d}x$ and thereby deduce *Parseval's formula*:

$$\frac{1}{2L}\int_{-L}^{L} |f(x)|^2\,\mathrm{d}x = \sum_{n\in\mathbb{Z}} |c_n|^2.$$

Hint: first show that the complex exponentials are orthonormal: $\dfrac{1}{2L}\displaystyle\int_{-L}^{L} \mathrm{e}^{i\pi mx/L}\,\mathrm{e}^{-i\pi nx/L}\,\mathrm{d}x = \delta_{m,n}$. You may also assume that you can swap the order of sums and integrals with impunity (of course you can justify the swaps if you like).

(c) Rewrite Parseval's formula in terms of the coefficients of the *real* Fourier series.

(d) Check your real version of Parseval's formula by applying it to the Fourier series for x worked out in lectures. You should obtain something equivalent to

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

studocu

Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

(e) Apply your Parseval formula to the Fourier series of x^2, $x^3 - x$ and $x^4 - 2x^2$ (maybe you can work them out quickly). You should recover evaluations of

$$\sum_{n=1}^{\infty} \frac{1}{n^4}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^6} \qquad \text{and} \qquad \sum_{n=1}^{\infty} \frac{1}{n^8}.$$

[By the way, $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^3}$ is not expected to be a rational multiple of π^3, though nobody seems to have proven this.]

Question 18.

Use separation of variables or one of the variations discussed in lectures to find a formal solution the heat equation for $u(x,t)$, with $x \in [0,1]$ and $t > 0$, where the following boundary and initial conditions are imposed:

$\partial_x u(0,t) = 0,$	$\partial_x u(1,t) = 0,$	$u(x,0) = 1 - x^2,$
$u(0,t) = 0,$	$\partial_x u(1,t) = 0,$	$u(x,0) = x(1 - \frac{1}{2}x),$
$\partial_x u(0,t) = 0,$	$\partial_x u(1,t) = 0,$	$u(x,0) = x^2(1 - \frac{2}{3}x),$
$\partial_x u(0,t) = 0,$	$u(1,t) = 0,$	$u(x,0) = 1 - x^2,$
$u(0,t) = \sin(\omega t),$	$u(1,t) = 0,$	$u(x,0) = 0,$
$u(0,t) = \cos(\omega t),$	$u(1,t) = 0,$	$u(x,0) = x(1 - x),$
$u(0,t) = 0,$	$\partial_x u(1,t) = t\,\mathrm{e}^{-t},$	$u(x,0) = x(1 - \frac{1}{2}x),$
$\partial_x u(0,t) = -\cos t,$	$u(1,t) = 2,$	$u(x,0) = 1 - x^2.$

Can you show that your formal solutions are genuine? Can you show that they are the unique solutions of these IBVPs?

Question 19.

Consider the (1D) inhomogeneous heat equation

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2} + h(x,t),$$

where $h(x,t)$ represents an external heat source (or sink). Find a formal solution of each of the following heat equation IBVPs with heat source h:

$u(0,t) = 0,$	$u(1,t) = 0,$	$u(x,0) = x(1 - x),$	$h(x,t) = \sin t,$
$\partial_x u(0,t) = 0,$	$u(1,t) = 0,$	$u(x,0) = 0,$	$h(x,t) = (1 - x^2)\cos t,$
$\partial_x u(0,t) = 0,$	$\partial_x u(1,t) = 0,$	$u(x,0) = 0,$	$h(x,t) = t\,\mathrm{e}^{-t}x^2(1 - \frac{2}{3}x),$
$u(0,t) = 5,$	$u(1,t) = 0,$	$u(x,0) = 0,$	$h(x,t) = x(1 - x)\sin t,$
$u(0,t) = 0,$	$\partial_x u(1,t) = \sin t,$	$u(x,0) = 0,$	$h(x,t) = x(1 - \frac{1}{2}x)\cos t.$

If you dare, investigate whether your formal solutions are genuine and/or unique.

Question 20.

Find formal solutions $u(x,t)$ to the (1D) wave equation, with $x \in [0,1]$ and $t \geqslant 0$, where the following boundary and initial conditions are imposed:

$u(0,t) = 0,$	$u(1,t) = 0,$	$u(x,0) = x(1 - x),$	$\partial_t u(x,0) = x(1 - x),$
$u(0,t) = 0,$	$\partial_x u(1,t) = 0,$	$u(x,0) = x(1 - \frac{1}{2}x),$	$\partial_t u(x,0) = x(1 - \frac{1}{2}x),$
$\partial_x u(0,t) = 0,$	$\partial_x u(1,t) = 0,$	$u(x,0) = 4x^3 - 6x^2 + 1,$	$\partial_t u(x,0) = 0,$
$\partial_x u(0,t) = \sin t,$	$u(1,t) = 0,$	$u(x,0) = 0,$	$\partial_t u(x,0) = 0.$

Repeat for the following variations on the wave equation:

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

- The *damped* wave equation: $\dfrac{\partial^2 u}{\partial t^2} = c^2 \dfrac{\partial^2 u}{\partial x^2} - \gamma \dfrac{\partial u}{\partial t}$, where $\gamma \geqslant 0$ is the *damping* coefficient.
- The *Klein-Gordon* equation of relativistic quantum mechanics: $\dfrac{\partial^2 u}{\partial t^2} = c^2 \dfrac{\partial^2 u}{\partial x^2} - \dfrac{m^2 c^4}{\hbar^2} u$, where m is the *mass*.
- The *telegraph* equation: $\dfrac{\partial^2 u}{\partial t^2} = c^2 \dfrac{\partial^2 u}{\partial x^2} - \gamma \dfrac{\partial u}{\partial t} - \zeta u$, where $\zeta \geqslant 0$ (see wikipedia).

Plot your solutions using a computer, choosing appropriate values for the various constants in each case. When your solution appears to be smooth, try to show that it is a genuine solution.

Question 21.

Consider the (1D) inhomogeneous wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + h(x,t),$$

where $h(x,t)$ accounts for external forces acting on the system. Find a formal solution to each of the following wave equation IBVPs with given inhomogeneous term h, plotting solutions and discussing whether the solution is genuine when appropriate:

$$u(0,t) = 0, \quad u(1,t) = 0, \quad u(x,0) = x(1-x), \qquad \partial_t u(x,0) = x, \quad h(x,t) = x \sin t,$$

$$u(0,t) = 0, \quad u(1,t) = 0, \quad u(x,0) = \begin{cases} \frac{x}{5} & \text{if } 0 \leqslant x \leqslant \frac{1}{2}, \\ \frac{1-x}{5} & \text{if } \frac{1}{2} \leqslant x \leqslant 1, \end{cases} \quad \partial_t u(x,0) = x, \quad h(x,t) = 10x(1-x)\cos(\omega t).$$

For the last IBVP, investigate the resonance at $\omega = 3\pi$ by, first, repeating the entire calculation for this value and, second, by carefully taking the limit $\omega \to 3\pi$ in your original solution.

Question 22.

Find formal solutions $u(x,t)$ to the (2D) Laplace equation, with $x, y \in [0,1]$, where the following boundary conditions are imposed:

$$\begin{array}{llll} u(0,y) = y(1-y), & u(1,y) = 0, & \partial_y u(x,0) = 0, & \partial_y u(x,1) = 0, \\ u(0,y) = 0, & u(1,y) = y^2 - y^3, & u(x,0) = 0, & \partial_y u(x,1) = 0, \\ u(0,y) = y(1-y), & \partial_x u(1,y) = 0, & u(x,0) = 0, & u(x,1) = x(1 - \frac{1}{2}x), \\ u(0,y) = 0, & u(1,y) = 1 - y^2, & \partial_y u(x,0) = 0, & u(x,1) = x(1-x). \end{array}$$

Plot your solutions using a computer and try to show that they are genuine solutions.

Question 23.

Find formal solutions $u(x,t)$ to the (2D) Laplace equation, *in polar coordinates*, on the given domain D with the following boundary conditions:

$$\begin{array}{llll} D = \left\{(r,\theta) \in [0,1] \times [0, \frac{\pi}{3}]\right\}, & u(r,0) = 0, & u(r, \frac{\pi}{3}) = 0, & u(1,\theta) = \theta(\frac{\pi}{3} - \theta), \\ D = \left\{(r,\theta) \in [0,1] \times [0, \frac{\pi}{2}]\right\}, & \partial_\theta u(r,0) = 0, & \partial_\theta u(r, \frac{\pi}{2}) = 0, & u(1,\theta) = \theta(1 - \dfrac{3}{2\pi}\theta), \\ D = \left\{(r,\theta) \in [0,1] \times [0, \frac{\pi}{4}]\right\}, & \partial_\theta u(r,0) = 0, & u(r, \frac{\pi}{4}) = 0, & u(1,\theta) = 1 - \dfrac{9}{\pi^2}\theta. \end{array}$$

Question 24.

Use the known transform

$$(\mathscr{F} \mathrm{e}^{-|x|})(p) = \frac{2}{1+p^2}$$

studocu

Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

and Fourier inversion to evaluate the following integrals:

$$\int_0^\infty \frac{\mathrm{d}p}{1+p^2}, \qquad \int_0^\infty \frac{\cos p}{1+p^2}\mathrm{d}p, \qquad \int_0^\infty \frac{p\sin p}{1+p^2}\mathrm{d}p.$$

Question 25.

Compute the Fourier transforms of the truncated waves

$$c(x) = \begin{cases} \cos(\alpha x) & \text{if } -a \leqslant x \leqslant a, \\ 0 & \text{otherwise,} \end{cases} \qquad s(x) = \begin{cases} \sin(\alpha x) & \text{if } -a \leqslant x \leqslant a, \\ 0 & \text{otherwise} \end{cases}$$

directly from the definition. Now compute them again using convolution and Heaviside functions. Are your answers consistent with the rules relating Fourier transforms and derivatives?

Question 26.

The definitions we have given for convolution in the Fourier and Laplace transform sections *do not agree*. The reason is that Laplace transforms are only defined on functions whose domain is $[0,\infty)$. Given any function $f(t)$, defined for all $t \in \mathbb{R}$, we can replace it by $\widetilde{f}(t) = f(t)H(t)$, so that its values for $t < 0$ are all 0. Show that the convolution of $\widetilde{f}(t)$ and $\widetilde{g}(t)$, using the definition given in the Fourier section, reproduces the convolution of f and g, using the definition given in the Laplace section.

Question 27.

Solve the following IBVPs for $x \in \mathbb{R}$ and $t \geqslant 0$, subject to the "boundary conditions" that the Fourier transform of the solution $u(x,t)$ exists, *ie.* $\int_{-\infty}^{\infty}|u(x,t)|\,\mathrm{d}x < \infty$ for all $t \geqslant 0$:

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2}, \quad u(x,0) = \mathrm{e}^{-x^2}; \qquad \frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}, \quad u(x,0) = 0, \quad \partial_t u(x,0) = x\mathrm{e}^{-|x|/2};$$

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2}, \quad u(x,0) = 20x\mathrm{e}^{-x^2}; \qquad \frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}, \quad u(x,0) = 2\operatorname{sech}^2 x, \quad \partial_t u(x,0) = 0;$$

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2}, \quad u(x,0) = \frac{1}{1+x^2}; \qquad \frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}, \quad u(x,0) = \frac{x}{x^2+16}, \quad \partial_t u(x,0) = \mathrm{e}^{-x^2/4}.$$

Plot your solutions on a computer and comment on their behaviour (convergence) as $t \to \infty$.

Question 28.

Use Fourier transform methods to solve the (2D) Laplace equation on the given domains D, subject to the given boundary conditions, and plot your solutions using a computer:

$$D = \{(x,y) \in \mathbb{R}\times[0,\infty)\}, \qquad u(x,0) = H_{-1}(x) - H_1(x), \qquad \int_0^\infty\int_{-\infty}^{\infty}|u(x,y)|\,\mathrm{d}x\mathrm{d}y < \infty,$$

$$D = \{(x,y) \in [0,1]\times\mathbb{R}\}, \qquad u(0,y) = 0, \quad u(1,y) = \delta_0(y).$$

For the last problem, you may need to use the decidedly non-trivial fact that

$$(\mathscr{F}_y\frac{\sinh(xy)}{\sinh(y)})(p) = \frac{\pi\sin(\pi x)}{\cos(\pi x)+\cosh(\pi p)}.$$

studocu

Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)